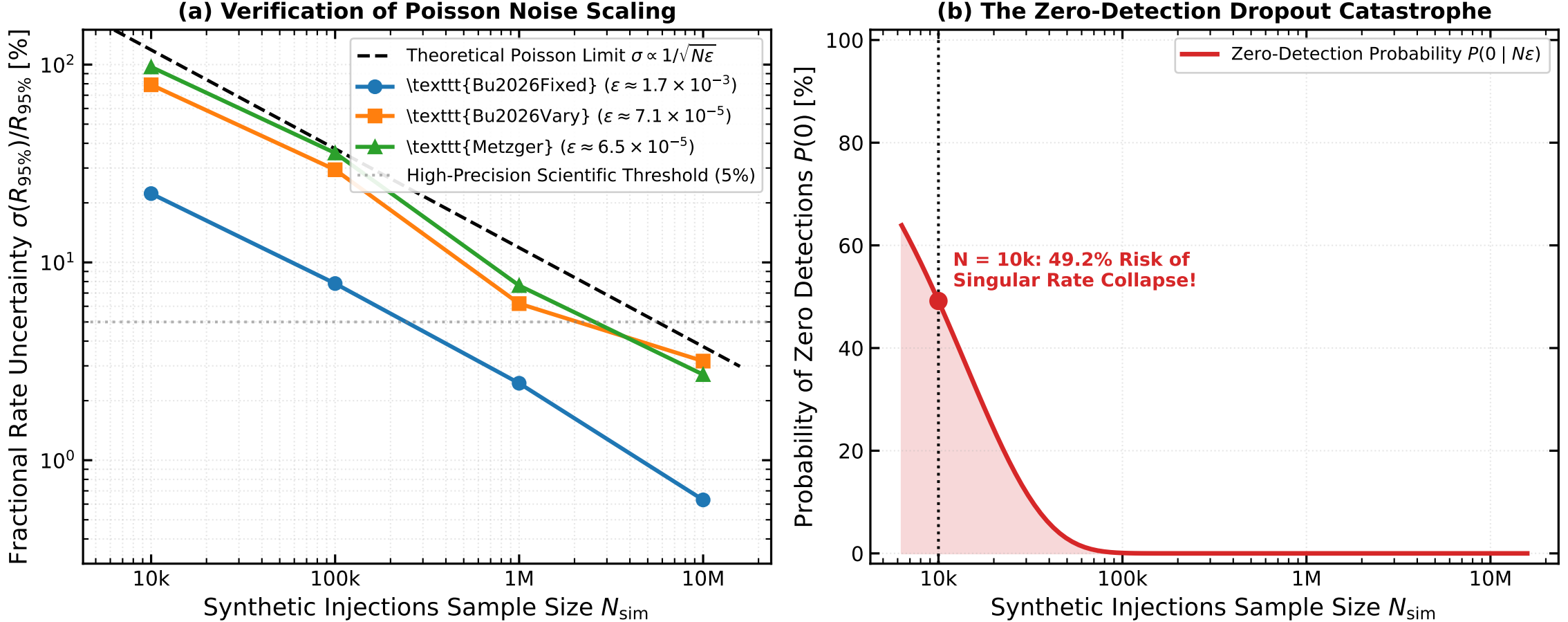
(a) Verification of Poisson Noise Scaling
Fractional Rate Uncertainty $\sigma(R_{95\%})/R_{95\%}$ [%]
10^2
10^1
10^0
10k
100k
1M
10M
Synthetic Injections Sample Size N_{sim}
Theoretical Poisson Limit $\sigma \propto 1/\sqrt{N\epsilon}$
\texttt{Bu2026Fixed} ($\epsilon \approx 1.7 \times 10^{-3}$)
\texttt{Bu2026Vary} ($\epsilon \approx 7.1 \times 10^{-5}$)
\texttt{Metzger} ($\epsilon \approx 6.5 \times 10^{-5}$)
High-Precision Scientific Threshold (5%)
(b) The Zero-Detection Dropout Catastrophe
Probability of Zero Detections $P(0)$ [%]
100
80
60
40
20
0
10k
100k
1M
10M
Synthetic Injections Sample Size N_{sim}
Zero-Detection Probability $P(0 \mid N\epsilon)$
N = 10k: 49.2% Risk of
Singular Rate Collapse!